

|   |   |                                                                                                                                                                                                                                                                                                                                                        |
|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   | Additional force required = $268.71 - (80.0)(9.81)\sin 5^\circ = 200 \text{ N (2 s.f.)}$                                                                                                                                                                                                                                                               |
| 4 | B | $x_1 = 10.0 - 7.5 = 2.5 \text{ cm}$<br>At constant speed:<br>$mg = kx_1$<br>$mg = kx_1 - (1)$<br>When slowing down:<br>$mg - kx_2 = ma - (2)$<br>Sub(1) into (2)<br>$kx_1 - kx_2 = ma - (3)$<br>$\frac{(3)}{(1)} \rightarrow \frac{a}{g} = \frac{x_1 - x_2}{x_1}$<br>$x_2 = 1.99 \approx 2.0 \text{ cm}$<br>new length = $10.0 - 2.0 = 8.0 \text{ cm}$ |
| 5 | B | Applying cosine rule                                                                                                                                                                                                                                                                                                                                   |